

The Efficacy and Safety of a Radial Approach versus a Distal Radial Approach for Diagnostic Coronary Angiography and Percutaneous Coronary Intervention: A Systematic Review and Meta-analysis

Toshihide Izumida MD¹, Jun Watanabe MD, PhD^{2,3}, Ryo Yoshida MD³, Kazuhiko Kotani MD, PhD²

1. Division of Community Medicine, Kanazawa Medical University Himi Municipal Hospital, 1130 Kurakawa, Himi, Toyama 935-8531, Japan
2. Center for Community Medicine, Jichi Medical University, 3311-1 Yakushiji Shimotsuke City, Tochigi, 329-0498 Japan
3. Department of Internal Medicine, Iwami Hospital, 1029-2 Iwami-Town, Tottori 681-0003, Japan.

Short title: Radial approach vs distal radial approach: A meta-analysis

1.Introduction

Coronary angiography (CAG) is an invasive but essential part of the diagnosis and treatment for coronary artery disease (CAD). Annually, it is estimated that 1,016,000 inpatient diagnostic CAG and 480,000 inpatient percutaneous coronary intervention (PCI) procedures are performed in the United States. [1] In European countries, it is estimated that 4,500 diagnostic coronary angiograms per million people and 2,000 PCI procedures per million people are performed each year. [2] Interventional cardiologists gain access via a peripheral artery and the latest guidelines from the European Society of Cardiology, National Institutes for Health and Care Excellence, and American College of Cardiology/American Heart Association recommended the radial approach (RA) over the transfemoral, transbrachial, and transulnar approaches, because it is associated with a reduced risk of cardiac death, all-cause mortality, bleeding, and access site complications. [3-5]

The distal radial approach (DRA) was recently introduced, as this approach may have some potential advantages in comparison to RA. [6-7] Previous observational studies showed that the two approaches were associated with similar successful cannulation rates [8], while the rates of vascular complications in DRA, including radial artery occlusion (RAO) and radial artery spasm (RAS), were less frequent than RA. [9-16] The DRA is

assumed to be an alternative approach to the RA, but the efficacy of the two approaches has never been systematically reviewed and analyzed.

Therefore, the present study aimed to evaluate the efficacy and safety of the DRA in comparison to the RA. To achieve this aim, a systematic review and meta-analysis of only randomized-controlled trials (RCTs) were conducted to produce high-quality evidence that informs healthcare decisions concerning these two approaches.

2. Research question

P: Adult patients (18 years of age or older) of either gender undergoing diagnostic CA and PCI for CAD.

I: Distal radial approach.

C: Radial approach.

O:

Primary outcome

1. Successful cannulation rate

defined as completion of cannulation without cross-over to another access site, or as defined by practitioners.

2. Incidence rate of RAS

RAS was diagnosed by the angiographic evaluation of the radial artery.

3. Incidence rate of RAO

RAO was diagnosed by the absence of flow by color Doppler ultrasound.

Secondary outcome

1. Mean puncture numbers per a patient

2. Mean time for hemostasis.

3. All adverse events

3. Method

3.1 Protocol

We followed the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 for preparing this protocol. [17] Our protocol will be published in protocols.io (<https://www.protocols.io/>).

3.2 Inclusion criteria of the articles for the review

3.2.1 Type of studies

We will include individual RCTs that evaluate the efficacy and safety of radial approach versus distal radial approach for cardiac catheterization. All papers including published, unpublished articles, abstract of conference and letter will be included, regardless of language, country restrictions, or publication year. Non-RCTs will be excluded.

3.2.2 Study participants

Inclusion criteria will be adult patients (18 years of age or older) undergoing diagnostic CA and PCI for CAD.

Exclusion criteria will be patients with more than 7 Fr sheath (available on a commercial basis). [18]

3.2.3 Intervention

Distal radial approach is a method of puncturing distal radial arteries at the proximal part of the anatomical snuffbox or the first intermetacarpal space. After successful artery puncture, a guidewire is smoothly passed through the needle and used to guide the sheath through the artery. After introduction of the sheath, interventional cardiologists perform the diagnostic CA and PCI through the sheath. [19]

3.2.4 Control

Radial approach is a method of puncturing radial artery at the forearm, a few centimeters above the wrist joint. [20]

3.3 Type of outcomes

3.3.1 Primary outcomes

All primary outcomes will include the definition of the authors of original studies.

1. Successful cannulation rate

Definition: defined as completion of cannulation without cross-over to another access site, or as defined by practitioners.

Period: during the examination

2. Incidence rate of RAS

Definition: RAS was diagnosed by the angiographic evaluation of the radial artery.

Period: during examination period

3. Incidence rate of RAO

Definition: RAO was diagnosed by the absence of flow by color Doppler ultrasound.

Period: during follow up period

3.3.2 Secondary outcomes

1. Mean number of punctures per patient

Definition: number of punctures

Period: during examination period

2. Mean time for hemostasis.

Definition:

Success of hemostasis was defined as no bleeding or hematoma formation after release.

Total time from when the sheath was removed to when the success of hemostasis was confirmed. It will include the definition of the authors of original studies.

Period: during examination period

3.4 Search method

3.4.1 Electronic search

The following databases will be searched:

1. the Cochrane Central Register of Controlled Trials (CENTRAL);
2. MEDLINE via PubMed;
3. EMBASE via ProQuest Dialog;

See Appendix 1, 2, and 3 for the search strategies.

3.4.2 Other resources

We will also search the following databases for ongoing or unpublished trials:

1. the World Health Organization International Clinical Trials Platform Search Portal (ICTRP);
2. ClinicalTrials.gov;

See Appendix 4, 5 for the search strategies.

We will check the reference lists of studies, including international guidelines published by European Society of Cardiology, National Institute for Health and Care Excellence, and American College of Cardiology/American Heart Association [3,4,5] as well as the reference lists of eligible studies and articles citing eligible studies. We will ask the authors of original studies for unpublished or additional data.

3.5 Data collection and analysis

3.5.1 Selection of the studies

Two independent reviewers (TI and RY) will screen titles and abstracts, followed by the assessment of the eligibility based on the full text. We will contact original authors if relevant data is missing. Disagreements between the two reviewers will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (JW).

3.5.2 Data extraction and management

Two reviewers (TI and RY) will perform independent data extraction of the included studies using standardized data collection form. We will use a pre-checked form using 10 randomly selected studies.

The form will include the information on study design, study population, interventions and outcomes. Any disagreements will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (JW).

3.6 Assessment of risk of bias in included studies

Two reviewers (TI and RY) will evaluate the risk of bias independently using the Risk of Bias 2. [21] Disagreements between the two reviewers will be discussed, and if this fails, a third reviewer will act as an arbiter (JW).

3.7 Measures of treatment effects

We will pool the relative risk ratios and the 95% confidence intervals (CIs) for the following binary variables: cannulation success, RAS, and RAO.

We will pool the mean differences and the 95% CIs for the following continuous variables: mean number of punctures per patient and mean time for hemostasis.

We will summarize adverse events based on the definition by the original article, but we will not perform meta-analysis.

3.9 Handling of missing data

We will ask not-presented data to the original authors.

3.9.1 Missing outcomes

The intention-to-treat (ITT) analysis will be performed for all dichotomous data as much as possible. For continuous data, missing data will not be imputed based on the recommendation by Cochrane handbook. [22] Meta-analysis will be performed about the available data in the original study.

3.9.2 Missing statistics

When original studies only report standard error or p-value, we will calculate the standard deviation based on the method by Altman. [23] If these values are missing when we contact the authors, standard deviation will be calculated by CI and t-value based on the method by Cochrane handbook [22], or validated method. [24] Validity of these methods will be analyzed by sensitivity analysis.

3.10 Assessment of heterogeneity

The statistical heterogeneity will be evaluated by visual inspection of the forest plots and calculating the I²-statistic (I² values of 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity). [22] When there is substantial

heterogeneity ($I^2 > 50\%$), the reason of the heterogeneity will be assessed. Cochrane Chi-test (Q-test) will be performed for I^2 statistic, and P value less than 0.10 will be defined as statistically significant.

3.11 Assessment of reporting bias

We will search the clinical trial registry system (ClinicalTrials.gov and ICTRP) and will perform extensive literature search for unpublished trials. We will assess the potential publication bias by visual inspection of the funnel plot. Egger test will be performed as well. We will not conduct the funnel plot and Egger test when we find less than 10 trials or trials. [22]

3.12 Meta-analysis

Meta-analysis will be performed using Review Manager software (RevMan 5.4.1). We will use a random-effects model.

3.13 Subgroup analysis

To elucidate the influence of effect modifiers on results, we will evaluate the subgroup analyses of the primary outcomes on the following factors when sufficient data are available.

1. (For participants): the young- to middle-age group (under 65 years) versus the elderly group (65 years and over)
2. (For intervention): right-side approach versus left-side approach [24, 25] and diagnostic CA versus PCI

3.14 Sensitivity analysis

The following sensitivity analyses for the primary outcomes will be undertaken to assess whether the results of the review are robust to the decisions made during the review process.

1. Exclusion of studies using imputed statistics.
2. Only the participants who complete the study with complete data

4. Summary of findings table

Summary of findings table will be made for the following outcome based on the Cochrane handbook. [22]

We will include grading to evaluate the quality of evidence based on the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach for each Summary of findings table. [26]

1. Successful cannulation rates
2. Radial artery spasm
3. Radial artery occlusion
4. Number of punctures per patient
5. Mean time for hemostasis

5. Conflict of Interest

The authors declare no conflicts of interests.

References

1. Virani SS, Alonso A, Benjamin EJ, Bittencourt MS, Callaway CW, Carson AP, Chamberlain AM, Chang AR, Cheng S, Delling FN, Djousse L, Elkind MSV, Ferguson JF, Fornage M, Khan SS, Kissela BM, Knutson KL, Kwan TW, Lackland DT, Lewis TT, Lichtman JH, Longenecker CT, Loop MS, Lutsey PL, Martin SS, Matsushita K, Moran AE, Mussolino ME, Perak AM, Rosamond WD, Roth GA, Sampson UKA, Satou GM, Schroeder EB, Shah SH, Shay CM, Spartano NL, Stokes A, Tirschwell DL, VanWagner LB, Tsao CW; American Heart Association Council on Epidemiology and Prevention Statistics Committee and Stroke Statistics Subcommittee. Heart Disease and Stroke Statistics-2020 Update: A Report From the American Heart Association. *Circulation*. 2020 Mar 3;141(9):e139-e596. doi: 10.1161/CIR.0000000000000757. Epub 2020 Jan 29. PMID: 31992061
2. Timmis A, Townsend N, Gale CP, Torbica A, Lettino M, Petersen SE, Mossialos EA, Maggioni AP, Kazakiewicz D, May HT, De Smedt D, Flather M, Zuhlke L, Beltrame JF, Huculeci R, Tavazzi L, Hindricks G, Bax J, Casadei B, Achenbach S, Wright L, Vardas P; European Society of Cardiology. European Society of Cardiology: Cardiovascular Disease Statistics 2019. *Eur Heart J*. 2020 Jan 1;41(1):12-85. doi: 10.1093/eurheartj/ehz859. PMID: 3182000
3. Knuuti J, Wijns W, Saraste A, Capodanno D, Barbato E, Funck-Brentano C, Prescott E, Storey RF, Deaton C, Cuisset T, Agewall S, Dickstein K, Edvardsen T, Escaned J, Gersh BJ, Svitil P, Gilard M, Hasdai D, Hatala R, Mahfoud F, Masip J, Muneretto C, Valgimigli M, Achenbach S, Bax JJ; ESC Scientific Document Group. 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes. *Eur Heart J*. 2020 Jan 14;41(3):407-477. doi: 10.1093/eurheartj/ehz425. PMID: 31504439
4. NICE guidelines [NG185]. Acute Coronary Syndrome, 2020. <https://www.nice.org.uk/guidance/ng185> (Published date: 18 November 2020).
5. Mason PJ, Shah B, Tamis-Holland JE, Bittl JA, Cohen MG, Safirstein J, Drachman DE, Valle JA, Rhodes D, Gilchrist IC; American Heart Association Interventional Cardiovascular Care Committee of the Council on Clinical Cardiology; Council on Cardiovascular and Stroke Nursing; Council on Peripheral Vascular Disease; and Council on Genomic and Precision Medicine. An Update on Radial Artery Access and Best Practices for Transradial Coronary Angiography and Intervention in Acute Coronary Syndrome: A Scientific Statement From the American Heart Association. *Circ Cardiovasc Interv*. 2018 Sep;11(9):e000035. doi: 10.1161/HCV.0000000000000035. PMID: 30354598
6. Kiemeneij F. Left distal transradial access in the anatomical snuffbox for coronary angiography (IdTRA) and interventions (IdTRI). *EuroIntervention* 2017;13:851–7.
7. Cai G, Huang H, Li F, Shi G, Yu X, Yu L. Distal transradial access: a review of the feasibility and safety in cardiovascular angiography and intervention. *BMC Cardiovasc Disord*. 2020 Aug 5;20(1):356. doi: 10.1186/s12872-020-01625-8. PMID: 32758150

8. Pua U, Sim JZT, Quek LHH, Kwan J, Lim GHT, Huang IKH. Feasibility Study of "Snuffbox" Radial Access for Visceral Interventions. *J Vasc Interv Radiol*. 2018 Sep;29(9):1276-1280. doi: 10.1016/j.jvir.2018.05.002. PMID: 30146194
9. Babunashvili A. Novel distal transradial approach for coronary and peripheral interventions. *J Am Coll Cardiol*. 2018;72(13 Supplement):B323.
10. Uddin MJ, Hashem S, Momen A, Sarker S, Rahman AU, Hasan M, Saha BP, Mozumder MR, Shahriar MS. Right distal radial artery access for coronary intervention initial experience in Bangladesh. *J Am Coll Cardiol*. 2019;73(15 Supplement):S72.
11. Soydan E, Akın M. Coronary angiography using the left distal radial approach - An alternative site to conventional radial coronary angiography. *Anatol J Cardiol*. 2018, 19(4): 243–248. doi: [https://doi.org/10.14744/ Anatol J Cardiol.2018.59932](https://doi.org/10.14744/AnatolJCardiol.2018.59932). PMID: 29578203.
12. Valsecchi O, Vassileva A, Cereda AF, Canova P, Satogami K, Fiocca L, Guagliumi G. Early clinical experience with right and left distal Transradial access in the anatomical snuffbox in 52 consecutive patients. *J Invasive Cardiol*. 2018, 30(6):218–223. PMID: 29543187.
13. Kim Y, Ahn Y, Kim I, Lee DH, Kim MC, Sim DS, Hong YJ, Kim JH, Jeong MH. Feasibility of coronary angiography and percutaneous coronary intervention via left snuffbox approach. *Korean Circ J*. 2018, 48(12):1120–1130. doi: <https://doi.org/10.4070/kcj.2018.0181>. PMID: 30088362.
14. Yu W, Hu P, Wang S, Yao L, Wang H, Dou L, Lu M, Bo G, Yu X, Chen J, Chen C, Luo Y, Yang M, Dong Z, Huang S. Distal radial artery access in the anatomical snuffbox for coronary angiography and intervention: A single center experience. *Medicine (Baltimore)*. 2020 Jan;99(3):e18330. doi: 10.1097/MD.00000000000018330. PMID: 32011433
15. Aoi S, Htun WW, Freeo S, Lee S, Kyaw H, Alfaro V, Coppola J, Pancholy S, Kwan T. Distal transradial artery access in the anatomical snuffbox for coronary angiography as an alternative access site for faster hemostasis. *Catheter Cardiovasc Interv*. 2019 Nov 1;94(5):651-657. doi: 10.1002/ccd.28155. Epub 2019 Feb 24. PMID: 30801939
16. Vefalı V, Sarıçam E. The Comparison of Traditional Radial Access and Novel Distal Radial Access for Cardiac Catheterization. *Cardiovasc Revasc Med*. 2020 Apr;21(4):496-500. doi: 10.1016/j.carrev.2019.07.001. Epub 2019 Jul 5. PMID: 31303367
17. Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ*. 2015;349(jan02 1):g7647-g7647. doi:10.1136/bmj.g7647
18. Gasparini GL, Garbo R, Gagnor A, Oreglia J, Mazzarotto P. First prospective multicentre experience with left distal transradial approach for coronary chronic total occlusion interventions using a 7 Fr Glidesheath Slender. *EuroIntervention*. 2019;15:126-128. doi: 10.4244/EIJ-D-18-00648. PMID: 30277464

19. Al-Azizi KM, Lotfi AS. The distal left radial artery access for coronary angiography and intervention: A new era. *Cardiovasc Revasc Med*. 2018 Dec;19(8S):35-40. doi: 10.1016/j.carrev.2018.03.020. Epub 2018 Mar 26. PMID: 29628223
20. Campeau L. Percutaneous radial artery approach for coronary angiography. *Cathet Cardiovasc Diagn*. 1989 Jan;16(1):3-7. doi: 10.1002/ccd.1810160103. PMID: 2912567
21. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. August 2019;14898. doi:10.1136/bmj.14898
22. Higgins JPT, Thomas J, Chandler J, et al. *Cochrane Handbook for Systematic Reviews of Interventions* version 6.0 (updated July 2019). Cochrane, 2019.
23. Furukawa T a, Barbui C, Cipriani A, Brambilla P, Watanabe N. Imputing missing standard deviations in meta-analyses can provide accurate results. *J Clin Epidemiol*. 2006;59(1):7-10. doi:10.1016/j.jclinepi.2005.06.006
24. De Rosa S, Torella D, Caiazzo G, Giampà S, Indolfi C. Left radial access for percutaneous coronary procedures: from neglected to performer? A meta-analysis of 14 studies including 7,603 procedures. *Int J Cardiol*. 2014 Jan 15;171(1):66-72. doi: 10.1016/j.ijcard.2013.11.046. Epub 2013 Nov 23. PMID: 24331866
25. Shah RM, Patel D, Abbate A, Cowley MJ, Jovin IS. Comparison of transradial coronary procedures via right radial versus left radial artery approach: A meta-analysis. *Catheter Cardiovasc Interv*. 2016 Dec;88(7):1027-1033. doi: 10.1002/ccd.26519. Epub 2016 Apr 1. PMID: 27037544
26. Guyatt G, Oxman AD, Akl E a, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol*. 2011;64(4):383-394. doi:10.1016/j.jclinepi.2010.04.026

Appendix 1: CENTRAL search strategy

#1 [mh "Percutaneous Coronary Intervention"]
 #2 (PCI OR "percutaneous coronary intervention"):ti,ab
 #3 "percutaneous coronary revascularization":ti,ab
 #4 (balloon NEAR/3 angioplasty):ti,ab
 #5 atherectomy:ti,ab 362 件 #6 [mh "Coronary Angiography"]
 #7 "coronary angiography":ti,ab
 #8 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7
 #9 (distal radial NEAR/3 procedure):ti,ab
 #10 (distal transradial NEAR/3 procedure):ti,ab
 #11 (distal radial NEAR/3 access):ti,ab
 #12 (distal transradial NEAR/3 access):ti,ab
 #13 (distal radial NEAR/3 approach):ti,ab

#14 (distal transradial NEAR/3 approach):ti,ab
 #15 #9 OR #10 OR #11 OR #12 OR #13 OR #14
 #16 (snuffbox or snuff-box):ti,ab
 #17 plamar:ti,ab
 #18 #15 OR #16
 #19 #8 AND #18

Appendix 2: MEDLINE (via PubMed) search strategy

1 exp Percutaneous Coronary Intervention/
 2 (PCI or "percutaneous coronary intervention*").tw.
 3 "percutaneous coronary revascularization*".tw.
 4 (balloon adj3 angioplast*).tw.
 5 atherectom*.tw.
 6 exp Coronary Angiography/
 7 coronary angiography.tw.
 8 or/1-7
 9 (distal radial adj3 procedure\$).tw. or (distal transradial adj3 procedure\$).tw. or (distal radial adj3 access\$).tw. or
 (distal transradial adj3 access\$).tw. or (distal radial adj3 approach\$).tw. or (distal transradial adj3 approach\$).tw.
 10 (snuffbox or snuff-box).tw.
 11 plamar.tw
 12 or/9-11
 13 8 and 12

Appendix 3: EMBASE (via ProQuest Dialog) search strategy

S13 S12 AND S8
 S12 S11 OR S10 OR S9
 S11 ab(plamar) OR ti(plamar)
 S10 ab(snuffbox OR snuff-box) OR ti(snuffbox OR snuff-box)
 S9 (ab("distal radial" AND (procedure OR access OR approach)) OR ti("distal radial" AND (procedure OR
 access OR approach)) OR ab("distal transradial" AND (procedure OR access OR approach)) OR ti("distal
 transradial" AND (procedure OR access OR approach)))
 S8 S7 OR S6 OR S5 OR S4 OR S3 OR S2 OR S1
 S7 ab(coronary angiography) OR ti(coronary angiography)
 S6 (EMB.EXACT.EXPLODE("coronary angiography"))
 S5 ab(atherectomy) OR ti(atherectomy)
 S4 ab(balloon AND angioplasty) OR ti(balloon AND angioplasty)

S3 ab("percutaneous coronary revascularization") OR ti("percutaneous coronary revascularization")

S2 ab(PCI) OR ti(PCI) OR ab("percutaneous coronary intervention") OR ti("percutaneous coronary intervention")

S1 (EMB.EXACT.EXPLODE("percutaneous coronary intervention"))

Appendix 4: ICTRP search strategy

Condition; "Coronary angiography" OR " Percutaneous coronary intervention" OR "PCI"

Intervention; "distal radial" OR "distal transradial" OR "snuffbox" OR "Palmar"

Appendix 5: ClinicalTrials.gov search strategy

"distal radial" OR "distal transradial" OR "snuffbox" OR "Palmar"